

layout: post title: ITS2 clean-up date: '2024-05-15' categories: tags: [DNA, ITS2, Extractions, Protocols]

projects: Mcap DT 2023; POC 2023 spawning

This post details information on ITS2 clean-up for the my ambient Mcap developmental timeseries samples from 2023 and for the POC spawning experimental samples from 2023. Mcap 2023 github repo is [here](#) and POC github is [here](#). See Ariana's [post](#) about the ITS2 protocol for this project.

Samples

Samples (44 total samples) were amplified on [5/13/24](#) and [5/14/24](#), but there appeared to be some [primer dimer](#) present, as evidenced by bands ~100bp. Therefore, we decided to do a bead clean-up on all samples.

Equipment and materials

- [Kapa Pure Beads](#)
- Tris-HCl (10 mM)
- Molecular grade ethanol
- Magnetic stand
- PCR tubes
- Gel rig + materials

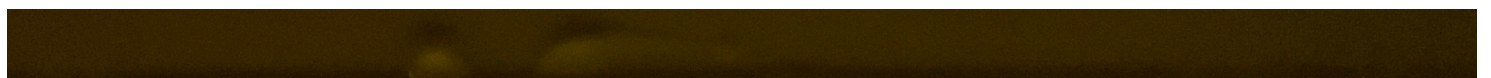
Protocol

Protocol was followed according to this [post](#). A few notes:

- After adding Kapa Beads, I incubated tubes for 15 minutes at room temperature.
- After the ethanol washes, I dried beads for ~4 minutes until beads were matte/ethanol had evaporated.
- After adding the elution buffer (10 mM Tris-HCl), I incubated tubes for 8 minutes at room temperature.

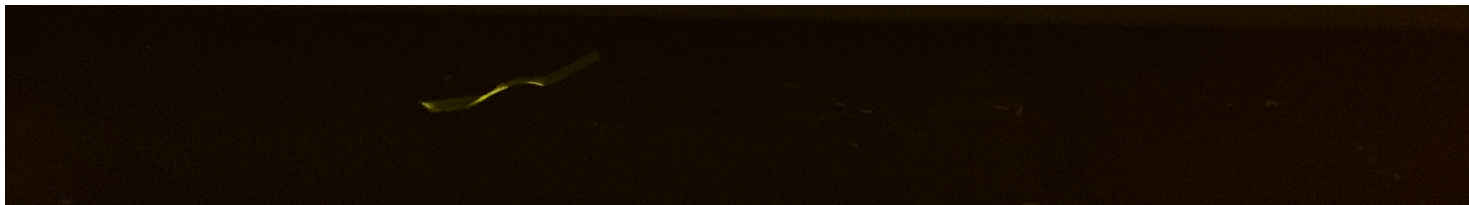
QC

Following the clean-up protocol, I ran a 2% gel for 90 minutes at 100 volts + 100 amps to check if the primer dimers were removed. I also used a 1kb ladder and a 100 bp ladder. The machine that the gel rig is hooked up to has only been reaching ~70-80 volts.



M48 M39 M73 M9 107 108 M51 M14 M60 8 M23 M72 M52 M8 110 M35 M63 M28 106 111 M6 M13 112 M83

M24 M36 POS M61 M74 M47 M86 M37 M26 9 6 M11 113 7 109 M87 POS NEG
NEG M88 M75 M62 M7 M10



Gel looks great! No primer dimer from what I can see. Hollie will send to Janet to confirm if these are good enough for sequencing.